

DOM & Events (Basic Understanding)

DOM stands for Document Object Model.

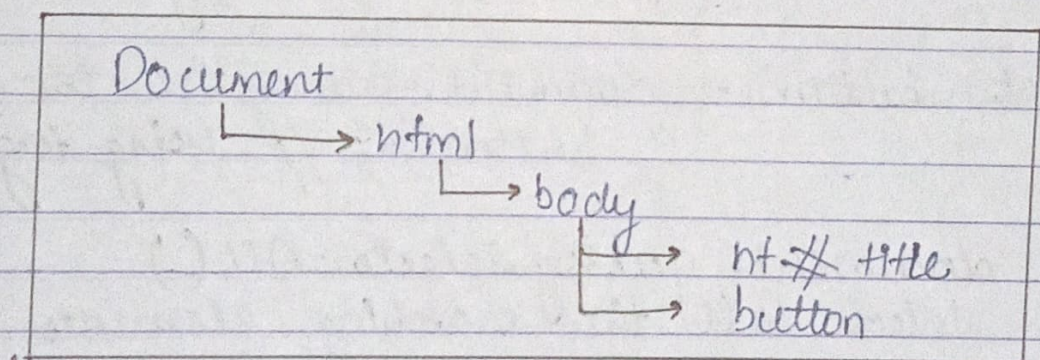
It's a tree like structure that represents your entire HTML webpage.

JS can use the DOM to view, change, add or delete elements dynamically.

Structure Example

```
<DOCTYPE html>
  <body>
    <h1 id="title"> Hello Users! </h1>
    <button> Click me! </button>
  </body>
</html>
```

The DOM tree looks like :-



Every HTML element (like `<h1>`, `<button>`) becomes a DOM node that JS can access.

Accessing / Selecting Elements

To work with the DOM, we must select the element we want to modify.

(*) `document.getElementById()`
Selects an element by its id.

```
let title =  
    document.getElementById("title");  
console.log(title.innerText);
```

// Output - Hello User!

(*) `document.querySelector()`
Selects the first element that matches a CSS selector.

```
let heading = document.querySelector("#title"  
    "#title"); // using ID
```

```
let button = document.querySelector(  
    "button"); // using tag name.
```

(*) `document.querySelectorAll()`
Selects all the matching elements.
(Returns a NodeList)


```
let buttons = document.querySelectorAll('button');  
console.log(buttons.length);  
// output - no. of buttons on page.
```

Changing Context and Style.

(*) Changing text:

```
title.innerText = "Welcome to MERN!";
```

(*) Changing HTML:

```
title.innerHTML = "<i> Hello Jee </i>";
```

(*) Changing styles:

```
title.style.color = "blue";
```

```
title.style.fontSize = "30px";
```

Adding or removing classes:-

```
title.classList.add("highlight");
```

```
title.classList.remove("old-style");
```


Event listener :

An Event is an action the user performs - like clicking a button, typing or scrolling.

The event listener allow JS to "listen" for these actions and respond

(*) Basic Syntax -

element.addEventListner ("event", function);

(*) Example - Click Event -

let button =

```
document.querySelector ("button");  
button.addEventListner ("click",  
    function () {  
        alert ("Button Clicked!");  
    }  
);
```

(*) Example - Mouse Over Event -

```
title.addEventListner ("mouseover",  
    function () {  
        title.style.color = "purple";  
    }  
);
```


Commonly Used Events

Events	Descriptions.
• Click	• triggered when an element is clicked
• mouseout	• when mouse leaves an element.
• mouseover	• when mouse pointer mouse over an element.
• keydown	• when a keyboard key is pressed.
• keyup	• when a key is released.
• input	• when a text input changes.
• submit	• when a form is submitted.
• load	• when the page or resource is fully loaded.

In Short

- DOM lets JS connect to HTML.
- Events let the user interact, with your webpage
- Together, they make websites dynamic & responsive - a must have skill for MERN developers.